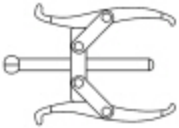


Drive Pinion Flange and Seal

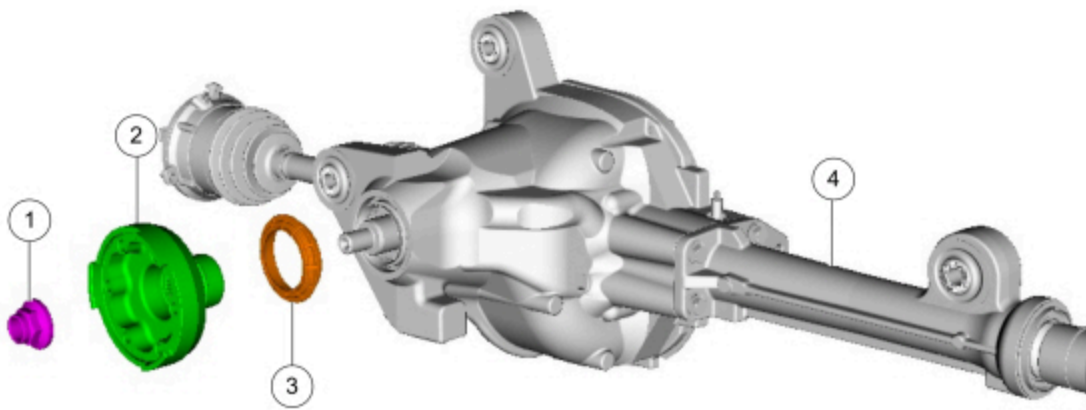


Special Tool(s)

 ST1321-A	2 Jaw Puller 205-D072 (D97L-4221-A) or equivalent
 ST1257-A	Holding Fixture, Drive Pinion Flange 205-126 (T78P-4851-A)
 ST1800-A	Installer, Drive Pinion Flange 205-002 (TOOL-4858-E)
 ST1358-A	Installer, Drive Pinion Oil Seal 205-208 (T83T-4676-A)

Material

Item	Specification
Motorcraft® SAE 80W-90 Premium Rear Axle Lubricant (US); Motorcraft® SAE 80W-90 Premium Axle Lubricant (Canada) XY-80W90-QL (US); CXY-80W90-1L (Canada)	WSP-M2C197-A



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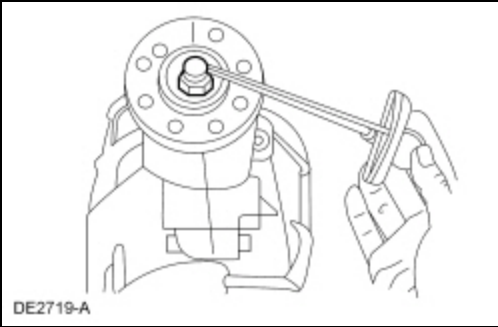
Item	Part Number	Description
1	389546 389546	Pinion nut and washer
2	4858 4858	Pinion flange
3	3N134 3N134	Drive pinion oil seal
4	3C202 3C202	Axle housing

Removal

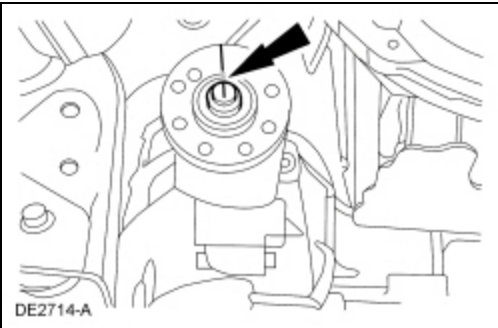
NOTICE: The color on the rear face of the drive pinion nut is critical to this repair. Use the same color new drive pinion nut for installation as the original. If a new collapsible spacer must be installed for pinion bearing preload reduction, install the nut supplied with the new spacer or damage to the component may occur.

NOTICE: This operation disturbs the pinion bearing preload. Carefully reset the pinion bearing preload during assembly.

1. With the vehicle in NEUTRAL, position it on a hoist. Refer to Section 100-02.
2. Remove the front brake pads. Refer to Section 206-03.
3. Remove the front driveshaft. Refer to Section 205-01.
4. Measure and record the pinion bearing preload.
 - Using a Nm (lb-in) torque wrench, rotate the pinion gear. Measure the torque required to maintain pinion gear rotation.

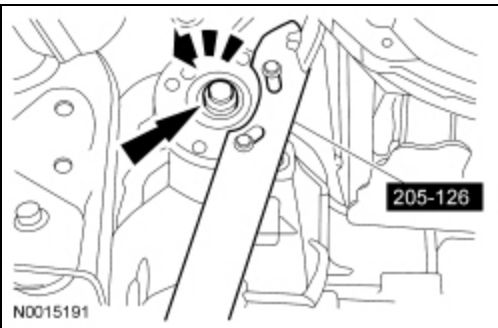


5. Index-mark the pinion flange and the drive pinion gear.

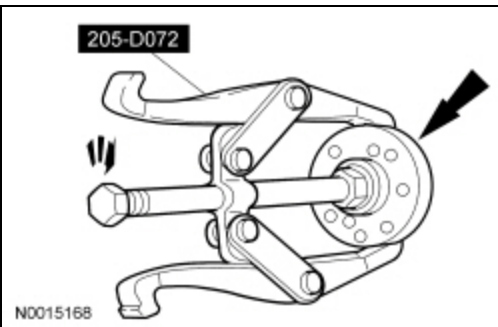


6. **NOTICE: Install a new pinion nut with the same color as the original if not replacing the collapsible spacer. If a new collapsible spacer is installed, install the nut in the kit or damage to the component may occur.**

Using the Drive Pinion Flange Holding Fixture to hold the pinion flange, remove and discard the pinion nut and washer.



7. Using the 2 Jaw Puller, remove the pinion flange.



8. Inspect the pinion flange for burrs and damage. Inspect the end of the pinion flange that contacts the pinion bearing cone, pinion nut counterbore and drive pinion seal surface for nicks. Discard the pinion flange if damaged.

9. **NOTICE: Do not damage the axle housing.**

Force up on the metal flange of the drive pinion seal. Install gripping pliers and strike with a hammer to remove the drive pinion seal.



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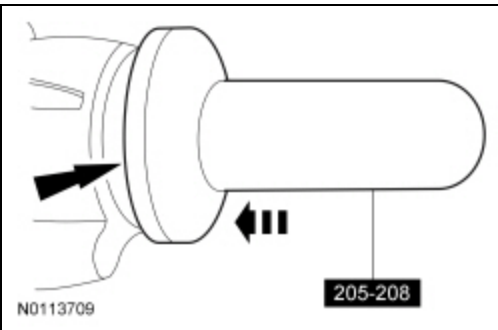
10. Verify the splines on the drive pinion gear are free of burrs. If burrs are evident, remove them with a fine crocus cloth.
11. Clean the drive pinion seal bore.

Installation

1. **NOTE:** Lubricate the drive pinion seal lips with axle lubricant.

NOTE: Drive pinion seal must be fully seated all the way around or drive pinion seal damage will occur.

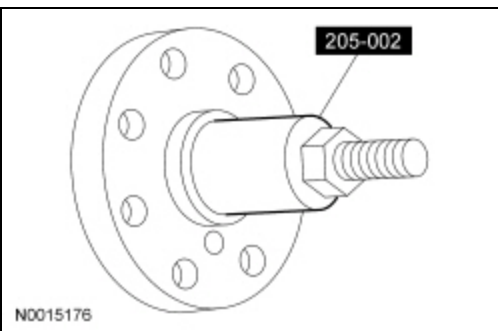
Using the Drive Pinion Oil Seal Installer, install the drive pinion seal.



2. **NOTICE:** Never use a hammer or install the pinion flange with power tools.

NOTE: Lubricate the pinion flange splines with axle lubricant.

Align the index marks made during removal and, using the Drive Pinion Flange Installer, install the pinion flange.



3. **NOTICE:** The color on the rear face of the drive pinion nut is critical to this repair. Use the same color new drive pinion nut for installation as the original. If a new collapsible spacer must be installed for pinion bearing preload reduction, install the nut supplied with the new spacer or damage to the component may occur.

Select the correct pinion nut for installation.

8.8 In. Drive Pinion Nut

Part Number	Color Description	Part Description
389546-S100	Orange	Pinion nut
4320	Purple	Pinion nut
4C121	Purple	Kit with spacer for all 223.52 mm (8.8 in) axles

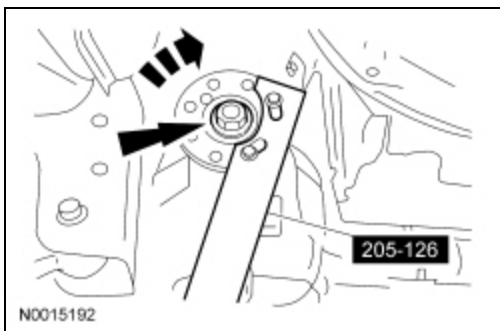
4. **NOTICE: Install a new pinion nut with the same color as the original if not replacing the collapsible spacer or damage to the component may occur.**

Install the new washer and pinion nut. Only hand-tighten the pinion nut at this time.

5. **NOTICE: Do not loosen the pinion nut to reduce drive pinion bearing preload. Install a new drive pinion collapsible spacer and pinion nut if drive pinion bearing preload reduction is necessary. If a new collapsible spacer must be installed for pinion bearing preload reduction, install the nut supplied with the new spacer or damage to the component may occur.**

Using the Drive Pinion Flange Holding Fixture to hold the pinion flange, tighten the pinion nut to set the drive pinion bearing preload.

- Tighten the pinion nut, rotating the pinion occasionally to make sure the drive pinion bearings are seating correctly. Take frequent drive pinion bearing preload readings by rotating the drive pinion gear with a Nm (lb-in) torque wrench. The final reading must be 0.56 Nm (5 lb-in) more than the initial reading taken during removal.



6. Install the front driveshaft. Refer to Section 205-01.

7. Install the disc brake pads. Refer to Section 206-03.